

1.2 Software and software development	
Types of software and the different methodologies used to develop software	
1.2.1 Systems Software	a) The function and purpose of operating systems. b) Memory Management (paging, segmentation and virtual memory). c) Interrupts. d) Scheduling: Round Robin, First come first served, Multi-level feedback queues, shortest job first and shortest remaining time. e) Distributed, Embedded, Multi-Tasking, Multi-User and Real Time operating systems. f) BIOS. g) Device drivers. h) Virtual machines.
1.2.2 Applications Generation	a) The nature of applications. b) Utilities. c) Open source vs Closed source. d) Translators: Interpreters, compilers and assemblers. e) Stages of compilation (Lexical Analysis, Syntax Analysis, Code Generation and Optimisation). f) Linkers and loaders.
1.2.3 Software Development	a) Understand the waterfall lifecycle, agile methodologies, extreme programming, the spiral model and rapid application development. b) Writing and following algorithms. c) The relative merits and drawbacks of different methodologies and when they might be used.
1.2.4 Types of Programming Language	a) Procedural languages. b) Assembly language (including following and writing simple programs with the Little Man Computer instruction set). c) Modes of addressing memory (immediate, direct, indirect and indexed). d) Object-oriented languages (using Java/C++ style pseudocode) with an understanding of classes, objects, methods, attributes, inheritance, encapsulation and polymorphism.